

Shadows in the Interface: A Comprehensive Study on Dark Patterns

LIMING NIE, Shenzhen Technology University, China

YANGYANG ZHAO*, Zhejiang Sci-Tech University, China

CHENGLIN LI*, Zhejiang Sci-Tech University, China

XUQIONG LUO, Changsha University of Science and Technology, China

YANG LIU, Nanyang Technological University, Singapore

As digital interfaces become increasingly prevalent, a series of ethical issues have surfaced, with dark patterns emerging as a key research focus. These manipulative design strategies are widely employed in User Interfaces (UI) with the primary aim of steering user behavior in favor of service providers, often at the expense of the users themselves. This paper aims to address three main challenges in the study of dark patterns: inconsistencies and incompleteness in classification, limitations of detection tools, and inadequacies in data comprehensiveness. In this paper, we introduce a comprehensive framework, called the Dark Pattern Analysis Framework (DPAF). Using this framework, we construct a comprehensive taxonomy of dark patterns, encompassing 64 types, each labeled with its impact on users and the likely scenarios in which it appears, validated through an industry survey. When assessing the capabilities of the detection tools and the completeness of the dataset, we find that of all dark patterns, the five detection tools can only identify 32, yielding a coverage rate of merely 50%. Although the four existing datasets collectively contain 5,566 instances, they cover only 32 of all types of dark patterns, also resulting in a total coverage rate of 50%. The results discussed above suggest that there is still significant room for advancement in the field of dark pattern detection. Through this research, we not only deepen our understanding of dark pattern classification and detection tools, but also offer valuable insights for future research and practice in this field.

CCS Concepts: • **Human-centered computing** → *HCI theory, concepts and models; Empirical studies in HCI.*

Additional Key Words and Phrases: Dark Pattern, Taxonomy, Detection tools, GUI

ACM Reference Format:

Liming Nie, Yangyang Zhao, Chenglin Li, Xuqiong Luo, and Yang Liu. 2024. Shadows in the Interface: A Comprehensive Study on Dark Patterns. *Proc. ACM Softw. Eng.* 1, FSE, Article 10 (July 2024), 22 pages. <https://doi.org/10.1145/3643736>

1 INTRODUCTION

The emergence of digital interfaces has fundamentally transformed the way we interact with technology, raising a series of ethical issues. Among these, "dark patterns" stand out as a significant concern [18, 31, 34]. These manipulative design strategies pervade User Interfaces (UI), with the aim of steering user behavior towards advantageous outcomes for service providers, often at the expense of users [42, 49, 57]. Given their widespread presence and impact on user choices and

*Corresponding authors.

Authors' addresses: Liming Nie, Shenzhen Technology University, Shenzhen, China, nieliming@sztu.edu.cn; Yangyang Zhao, Zhejiang Sci-Tech University, Hangzhou, China, yangyangzhao@zstu.edu.cn; Chenglin Li, Zhejiang Sci-Tech University, Hangzhou, China, lichenglin2110@163.com; Xuqiong Luo, Changsha University of Science and Technology, Changsha, China, luoxuqiong@csust.edu.cn; Yang Liu, Nanyang Technological University, Singapore, Singapore, yangliu@ntu.edu.sg.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

© 2024 Copyright held by the owner/author(s).

ACM 2994-970X/2024/7-ART10

<https://doi.org/10.1145/3643736>

privacy, dark patterns have attracted substantial regulatory attention [14, 25, 43, 51]. However, attempts to address this issue are hindered by the absence of a comprehensive framework for understanding and classifying them, underlining the importance of establishing a robust taxonomy and analysis framework.

Dark patterns have increasingly become the focus of academic research due to their increasing ethical and practical implications across various digital platforms [32]. The seminal work of Harry Brignull in 2010 laid the foundations for this field [17], offering an initial classification system. Current research primarily focuses on two areas: classification and detection. The field of classification has evolved significantly since its inception, moving from simplistic taxonomies to more nuanced categories that consider various digital platforms and cognitive biases exploited [29, 41]. Despite this, there is a continual effort to develop a universally accepted taxonomy that encompasses newly identified dark patterns [23, 31, 50]. Similarly, detection methods have evolved [20, 38, 48]. The initial approaches relied heavily on manual scrutiny, which, although accurate, is impractical due to its time-consuming nature [37]. Subsequent semi-automated methods employed clustering techniques to mimic user behavior [50]. Recent work has explored machine learning approaches for specific contexts such as cookie banners but requires labeled training data [48].

However, current research faces several limitations: The first is that the prior taxonomies are inconsistent and incomplete. Most studies are confined to specific types or contexts and lack a comprehensive taxonomy. Meanwhile, existing taxonomies often overlook user impact and likely scenarios where dark patterns appear. The second is the limitations of the detection tools. Existing tools often suffer from incomplete coverage or low accuracy. No study has yet analyzed the full range of the capabilities of these tools in a complete taxonomy. The third is the completeness of the data. Previous research often relies on limited datasets, which may not be representative, and there is a notable absence of holistic analysis on their coverage and utility.

In this paper, we introduce the Dark Pattern Analysis Framework (DPAF), a novel two-stage approach aimed at addressing existing gaps in the field. Initially, the framework employs a systematic literature review to assess the current state of dark pattern taxonomies. Subsequently, it embarks on taxonomy construction and labeling. Specifically, we conduct a relatively comprehensive taxonomy of dark patterns that not only integrates existing taxonomies, but also adds the types not included in previous taxonomies. In addition, each type is labeled with its impact on users and the likely scenarios in which it appears. Following this, we evaluate the capabilities of existing dark pattern detection tools and examine the extent to which existing datasets encompass the types of dark patterns identified by the newly constructed taxonomy. Through this approach, our aim is to answer three research questions (RQ):

- **RQ1:** Are current dark pattern taxonomies comprehensive? Our systematic review highlights the incompleteness of existing dark pattern taxonomies. To fill this gap, we have created the most exhaustive dark pattern taxonomy to date, comprising 64 unique types, validated through an industry survey. We have also labeled each type based on its impact on users and the contexts in which it is likely to appear, enhancing the taxonomy's applicability.
- **RQ2:** What are the capabilities and limitations of current Dark Pattern detection tools? Of the 64 types of dark patterns, the five selected tools could identify only 32, with a coverage rate of only 50%. Furthermore, these tools do not identify the remaining 32 patterns, which constitute 50% of the total.
- **RQ3:** What is the current status of the available data supporting automated Dark Pattern recognition? The four available datasets contain 5,566 instances, covering only 32 of the 64 types of dark patterns, with a coverage rate of 50%. The absence of instances for the remaining 32 types limits the capabilities of current detection tools.

By answering these questions, this paper deepens our understanding of dark pattern classification and detection tools, offering valuable insights for the field's future research and practice. For further details, please consult our online website [7]. Our contributions are as follows:

- Conducted the first systematic literature review focused solely on dark patterns.
- Established a comprehensive dark pattern taxonomy, fostering standardized communication among various stakeholders.
- Identified current research limitations and suggested directions for future studies.

2 BACKGROUND AND MOTIVATION

2.1 Dark Patterns

In contemporary society, User Interface (UI) and User Experience (UX) design have become deeply embedded in our daily lives. Whether shopping, socializing, or working, we constantly interact with various applications and online platforms. The design of these platforms directly influences our decisions, emotions, and behaviors. An adeptly designed interface can streamline tasks, enhance user experience, and even nudge consumers to make wiser choices. However, with technological advancements and intensifying business competition, a design strategy called "dark patterns" has gradually emerged.

Dark patterns, a term introduced in the scholarly literature [13, 19], refer to manipulative or deceptive elements in UX design. These are deliberately constructed designs with the intent of inducing users to take actions they might not willingly choose if they were fully informed or aware of alternatives [50]. This strategy capitalizes on individuals' cognitive biases and habitual behaviors to achieve specific commercial or other objectives, such as enticing users to mistakenly click on advertisements, unknowingly agree to subscriptions, or share personal data without a complete understanding. Dark patterns show themselves in multiple forms, such as misleading buttons, hidden fees at checkout, complex unsubscribe procedures, and ads disguised as authentic content. Such design practices may erode trust in products or services and, at times, might even incur legal liabilities, posing potential adverse impacts on users, businesses, and the broader online ecosystem [26, 61]. Fundamentally, dark patterns are considered unethical design tactics because they manipulate and deceive users, coercing them into choices that benefit on-line services, frequently to the detriment of the users' best interests.

2.2 Motivation

The increasing prevalence of dark patterns in user interfaces is raising significant concerns. These deceptive designs can undermine user trust, nudging them towards decisions that are not in their best interest, potentially diminishing user engagement and retention. Beyond mere manipulation, dark patterns can coerce users into unintentional purchases or the sharing of sensitive information, posing legal challenges to companies and violating user rights. Furthermore, by instilling a sense of urgency, these patterns increase psychological stress, which could evolve into wider mental health problems. From the standpoint of data security, dark patterns increase the risk of data breaches when users are duped into providing too much information. Although dark patterns might improve short-term metrics, the extensive and adverse long-term effects on both individuals and society warrant serious consideration.

The widespread use of dark patterns and their potential negative effects highlight an urgent need to better understand and identify these deceptive practices. This is crucial to preventing their development and improving the overall quality of software. However, tackling dark patterns presents numerous challenges in technical, conceptual, and practical aspects. Key obstacles include:

- *Inconsistencies and Incompleteness in Classification*: Although significant research effort has been aimed at creating classification frameworks for dark patterns, much of this work is limited to certain types or contexts, without a unified and standardized categorization. An all-encompassing taxonomy that includes dark patterns from various sources has yet to be established. This lack of standardization hampers efforts to investigate, identify, and address these manipulative design practices.
- *Limitations of Detection Tools*: Given the potential hazards posed by dark patterns, the creation and implementation of automated detection tools are crucial. These tools can serve as a deterrent, encouraging designers and developers to follow ethical and user-focused design principles. However, the detection effectiveness of current tools varies, and many struggle to achieve thorough coverage or maintain accuracy. Critical assessment of the capabilities and limitations of these tools is vital for fostering further investigation and improvement in this area. Despite the clear need for such evaluations, research dedicated to examining these aspects has been conspicuously absent to date.
- *Inadequacies in Data Comprehensiveness*: The effectiveness of dark pattern detection tools is significantly influenced by the quality and size of the datasets they rely on. Inadequacies or biases in these datasets can diminish the effectiveness of the tools. Numerous studies have used limited datasets that may not fully represent the broader context, with a notable absence of comprehensive analysis on their inclusion and applicability. Therefore, it is essential to evaluate current datasets, focusing on their diversity and thoroughness.

In light of these existing challenges, a comprehensive exploration of dark patterns becomes imperative. Motivated by this urgency, our research seeks to bridge the existing knowledge gaps, offering a robust unified classification standard and a detailed examination of the current dark pattern detection tools and available data. Through our endeavors, we strive to advance both academic discourse and practical interventions related to dark patterns.

3 METHODOLOGY

In this section, we first address three research questions and the intentions behind them. Subsequently, the overall framework of the corresponding empirical study is introduced. Following that, we will outline each segment and step within the proposed framework in detail, especially focusing on the construction of the dark pattern taxonomy, annotation of tool capabilities and limitations, and the process of marking data availability.

3.1 Research Questions(RQs)

- RQ1: Are current dark pattern taxonomies comprehensive? To effectively understand, detect, and intervene with dark patterns, a comprehensive and widely accepted taxonomy is needed. A clear and comprehensive taxonomy can help researchers and relevant practitioners better grasp the diversity and complexity of dark patterns. It can strongly support the formulation of related policies and regulations and can also help improve the capabilities of related detection tools. Therefore, building the most comprehensive and industry-accepted dark pattern taxonomy is of paramount importance.
- RQ2: What are the capabilities and limitations of current Dark Pattern detection tools? Identification of dark patterns is time-consuming and requires specialized expert experience. Although some automated detection tools have been proposed, the capabilities and limitations of these detection tools have not been adequately researched or understood. Therefore, a thorough investigation of these aspects is crucial for further optimizing these tools, enhancing their accuracy and reliability, and expanding their scope of application.

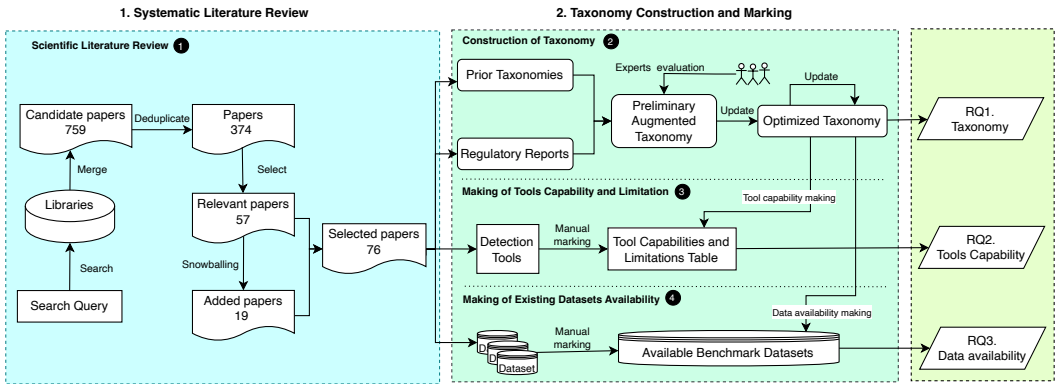


Fig. 1. The framework of DPAF

- RQ3: What is the current status of the available data supporting automated Dark Pattern recognition? A significant obstacle hindering research and solutions related to dark patterns is the lack of comprehensive and reliable datasets. However, the current status of data availability has not been adequately studied. Thus, a deep analysis of the availability of existing data and its actual coverage in the comprehensive dark pattern taxonomy is necessary and urgent.

3.2 Framework

To address the three research questions mentioned above, we propose a framework, named **Dark Pattern Analysis Framework (DPAF)**, based on the taxonomy construction method used by Ladisa et al. [45]. Our framework entails constructing a taxonomy for dark patterns, annotating the capabilities and limitations of existing detection tools within this taxonomy, marking the availability of data, and an update mechanism. As shown in Figure 1, DPAF comprises two stages: *Systematic Literature Review* and *Taxonomy Construction and Marking*. We will discuss each stage and the specific steps involved.

3.3 Scientific Literature Review

This section describes a systematic review of academic literature related to dark pattern research ①. To achieve this, we followed a structured methodology to perform a literature review, encompassing various stages [39, 54].

3.3.1 Design of Search Query. Our systematic review of dark pattern research began with the design of an effective search query. To ensure a robust query, we initiated an exploratory search on Google Scholar using the keyword "dark pattern." This preliminary step revealed four relevant studies [18, 40, 48, 50] that provided initial insight into the topic and helped us identify related keywords. Subsequently, we analyzed these keywords to construct a search query string that would yield the most relevant papers on dark patterns. The search query is as follows:

(Dark patterns OR dark nudging OR Digital sludging OR Manipulation techniques OR Deception techniques OR deceptive button OR dark pattern recognition OR dark pattern classification AND GUI OR UI OR user interface OR graphical user interface)

3.3.2 Literature Search. With our well-crafted search query in hand, we proceeded to conduct an extensive literature search across five reputable databases: IEEE Xplore, Scopus, Google Scholar,

ACM Digital Library, and arXiv. Specifically, we incorporated selected arXiv papers to ensure a comprehensive view of current research. We limited our search scope to titles, abstracts, and author keywords and focused exclusively on publications in the English language, including journals, conference proceedings, and book chapters.

3.3.3 Merge Candidate Literature. Following our systematic literature search, we recorded the number of candidate literature retrieved from each database. The breakdown was as follows: Google Scholar (461), IEEE Xplore (107), Scopus (83), ACM Digital Library (91), and arXiv (17). In total, we initially identified 759 candidate papers.

3.3.4 Deduplication. To ensure the integrity of our review, we meticulously removed duplicate entries from the pool of candidate literature. This process resulted in a refined set of 374 papers that were subsequently subjected to the next stage of paper selection based on inclusion criteria.

3.3.5 Paper Selection. The paper selection process was carried out in two phases. During the initial phase, we conducted a preselection by scrutinizing the meta-information of each paper, specifically focusing on publication type, title, and abstract. Papers that were clearly irrelevant to dark patterns in the UIs were excluded during this phase. This initial screening reduced the candidate literature to 143 articles, which advanced to the second phase of selection. In the second phase, we acquired the full text of all remaining papers and conducted a comprehensive examination of their content. Specifically, we dived into the introduction, methods, and results sections to confirm whether each paper presented a study related to dark patterns in UIs. Furthermore, all sections of the study had to be written in English to be considered. Papers that did not meet these criteria were rejected. To improve the precision and consistency of our review, a parallel review strategy was adopted. Each piece of literature was independently reviewed by at least two authors. Inclusion in our final review required a consensus on the relevance of the article among the authors. At the end of this rigorous selection process, we identified 57 relevant papers.

3.3.6 Snowballing. To further enhance the completeness of our selection, we adopted a bidirectional snowballing strategy, following references both forward and backward from the 57 papers initially deemed relevant. This approach applied the established selection criteria consistently across all rounds of snowballing. We included only those documents that were directly cited or were cited by our chosen literature, capping the inclusion at a maximum of two levels of citation depth to maintain focus and manageability. This iterative process led to the inclusion of 19 additional new papers. Consequently, we arrive at a final selection of 76 relevant papers that formed the foundation of our comprehensive study of dark patterns.

Of the 76 papers on dark patterns, 7 originate from arXiv. Subsequent checks revealed that 4 are formally published, leaving 3 exclusive to arXiv. Refer to our website[7] for more information. Note that these 3 papers did not reveal new dark pattern types. Figure 2 shows the publishing trends per year according to the 76 selected papers. These publications were meticulously reviewed to extract details related to prevalent dark patterns and the recognition and classification of such patterns. These extracted data serve as the basis for answering our RQs.

3.4 Construction of Taxonomy

This subsection delineates the methodology for constructing an optimized taxonomy ②, as shown in Figure 1, which serves as an augmented version accepted by the industry. To achieve this aim, the process is divided into three sequential stages: First, we construct a *preliminary enhanced taxonomy* that serves as an enhanced version of existing taxonomies. Second, we conduct a survey to industry experts to assess the rationality, completeness, and usefulness of this preliminary taxonomy. Finally,

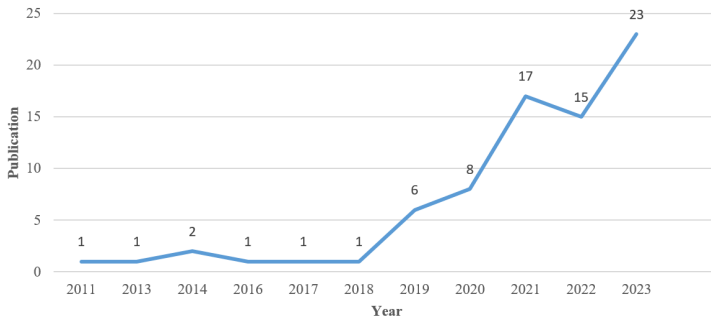


Fig. 2. Annual Publication Trend of Dark Patterns-Related Research Papers

based on the feedback received from these experts, the taxonomy is refined to produce the final *optimized taxonomy*.

3.4.1 Building an Preliminary Augmented Taxonomy. The main goal of this step is to build an augmented version of the taxonomy based on previous work on dark patterns. Previous studies mainly included prior taxonomies [15, 19, 31, 32, 46, 50], reports [10, 11, 21, 47, 55], and some dark patterns that appeared in the articles but were not included in prior taxonomies [20, 24, 30, 35, 44, 48, 61]. In addition to this, we annotate each type of dark pattern contained within our Preliminary Augmented Taxonomy in terms of its potential user impact and application scenarios. The objective is to gain a more comprehensive understanding of the potential effects of dark patterns on users, as well as their scope of application across various contexts. To achieve the objectives stated above, we follow a multistep approach divided into four principal phases.

A. Aggregation of Existing Taxonomies. First, from 76 related articles, we identified 15 key taxonomies that provide a substantive basis for our work. We manually extracted existing dark pattern types along with their classification information from the prior taxonomies.

Second, using the latest taxonomy proposed by Gray et al.[32] as a baseline, we consolidated similar dark patterns to avoid redundancy. This consolidation was classified into two categories: *Textual Similarity* and *Conceptual Similarity*. On the one hand, textual similarity refers to dark patterns that either have the exact same definition, point to the same dark pattern instance, or use synonyms. For these types of patterns, we only need to record their paper sources, descriptions, and the type attribution of these patterns in the original taxonomy version. For those whose categories are inconsistent, three of the authors will discuss and determine the final category based on existing classifications and the principle of majority submission. On the other hand, conceptual similarity means that patterns are different in terms but similar in nature. Because we do not know in advance which dark patterns are conceptually similar, we use manual confirmation to check all patterns, except textual similarity. Specifically, two of the authors will read each pattern separately. The text descriptions, related examples, and attribution information corresponding to the remaining unconfirmed patterns give you your own conclusion, that is, they either belong to the same dark pattern or should belong to a separate pattern. After the two authors completed all the annotations, find patterns with differences in the annotations. For these annotations, a third author was asked to make independent judgments and finally negotiated. The outcome of this phase was an initial version of the most comprehensive taxonomy that covers all known dark patterns.

Conceptual similarity is for more complex similar dark pattern judgments. It mainly includes the following steps: a). similarity judgment of description: for the definition of dark patterns, although they may be described by different words, they may express the same or similar meanings. These kinds of dark patterns can be merged into the same class of dark patterns. For all detection tools, the corpus is an important basis for detecting dark patterns. Therefore, this step is a way of

categorization in line with the detection basis. b). Judging by the similarity of the presentation of instances: most of the existing detection tools take visual cue as an important basis for detecting dark patterns. Therefore, we care whether the typical instances of the two types of dark pattern have similarity in their expressions. For example, the graphical interfaces when they appear and the descriptions. c). Judgment of Potential Impact on Users: In order to be consistent with the real-world discussion of dark patterns, we include the different types of impact that dark patterns may have on users as an additional judgmental step. The authors carefully reviewed five rules proposed by regulatory agencies or organizations. These articles made generalizations about dark patterns, and some of their comments were adopted by us in constructing the dark pattern taxonomy.

In addition, in previous dark pattern classification methods, almost all existing detection tools use the smallest unit of dark pattern type as a detection unit. Because they are widely discussed, they are retained as representative types during the construction of the taxonomy. This means that they are typically not merged into other dark pattern types. Given the importance of classification for detection tools, our website offers specific examples of merging to illustrate the process of combining dark pattern types more effectively.

B. Add Missing Dark Patterns. In building upon the initial taxonomy of dark patterns, which aggregated findings from prior literature, our objective is to construct the most comprehensive taxonomy to date. Despite the extensive categorization already provided in existing taxonomies, such as the seminal work by Gray et al. [32], we identified coverage gaps that may have resulted from divergent focal points in previous research. To rectify this, we undertook a deep reading of all relevant literature, comprising 76 papers, in order to broaden the scope of our taxonomy. Specifically, we have supplemented the latest taxonomy from [32] by incorporating additional 10 dark patterns, thereby establishing a preliminary taxonomy for the upcoming survey phase.

C. Assessment of User Impact. Although much of the prior research has primarily focused on the identification and description of dark patterns, there has been a comparatively limited exploration of the end impact these patterns impose on consumers. In an attempt to offer a comprehensive understanding of the multi-dimensional ramifications that all variants of dark patterns could potentially exert on users, we engage in a systematic annotation of each identified type of dark pattern with its associated harms. For the purposes of this study, we employ the foundational framework proposed by the Organization for Economic Cooperation and Development (OECD) to categorize the harms of dark patterns [55]. This framework identifies dark pattern harms from six unique vantage points, which, for ease of reference and clarity, we herein label as H1 to H6:

- *Harm to User Autonomy (H1):* This category suggests that certain dark patterns compromise user autonomy by forcing consumers to make choices they would not otherwise make, limiting the available options, and obfuscating the decision-making process.
- *Personal User Detriment:* This domain encompasses three separate, yet interconnected, harms: *Financial Loss (H2)*, *Privacy Harms (H3)*, and *Psychological Detriment and Time Loss (H4)*. H2 addresses dark patterns that manipulate consumers into making unnecessary purchases or overspending. H3 emphasizes how dark patterns could induce users into unintentionally disseminating excessive personal information, thereby elevating their risk exposure. H4 reflects on dark patterns that place emotional and cognitive strain on users, exploiting their vulnerabilities and potentially leading to time wastage.
- *Structural User Detriment:* This facet includes two distinct, yet interrelated, harms: *Weaker or Distorted Competition (H5)* and *Reduced Consumer Trust and Engagement (H6)*. H5 means that dark patterns can distort market competition by preventing or dissuading consumers from shopping and comparing offers. H6 posits that dark patterns may erode consumer trust and

engagement by deceiving users into oversharing information or overspending, particularly if such practices are subsequently uncovered.

In alignment with the aforesaid definitions, we performed a meticulous assessment of each identified dark pattern to gauge its potential impact on users. This process consisted of two pivotal phases: First, we unambiguously included those dark patterns that were explicitly cited by the OECD as emblematic instances of particular kinds of harm. These were directly incorporated into our analytical framework. Subsequently, for any unaccounted for dark patterns that were not explicitly delineated, our interdisciplinary team of three authors undertook rigorous collective discussions. These discussions aimed to systematically assign potential harms to users, guided by the definitions and corresponding examples established by the OECD [55].

D. Common Scenarios. To gain a more comprehensive understanding of the possible impacts of dark patterns on users, particularly in different application scenarios, it is imperative to annotate the scenarios in which these patterns manifest. Such annotation serves a dual purpose: it not only identifies the broad applications of dark patterns but also delves deeper into their potential impacts within diverse user groups and environments. Towards this aim, we have deployed a two-fold methodological approach for systematically annotating various dark pattern scenarios.

Initially, we amalgamated dark pattern scenarios discussed in previous research [37, 43] and categorized them into four distinct classes: *Social Platforms (S)*, *E-commerce (E)*, *Entertainment Software (ES)*, and *General or Universal (All)*. These classifications serve as a robust foundation for the subsequent annotation of dark patterns. In the second step, we embark on an intricate and detailed annotation process. Initially, we extract pertinent data regarding the potential impact of dark patterns across various scenarios from existing scholarly publications. To elaborate, suppose a particular paper concentrates on categorizing types of dark patterns prevalent on social platforms [59]. During our data acquisition phase, we would not only incorporate all types of dark patterns detailed in that paper into our taxonomy but also label them under the dimension of *Social Platform* to denote the scope of their impact. This approach might result in a specific type of dark pattern being discussed in multiple articles centered around different application scenarios, thus acquiring multiple tags. This is, in fact, an expected outcome. Our taxonomy specifically identifies three key application scenarios: *Social Platforms*, *E-commerce*, and *Entertainment Software*. Therefore, should a dark pattern type be prevalent across these three scenarios, we assign it the designation *All*.

At this point, we can complete the annotation of all dark pattern type impact scenarios, thereby furnishing a nuanced and comprehensive framework that enhances our understanding of how these patterns operate across various contexts.

Upon the successful completion of these key steps, we have established a taxonomy of existing dark patterns that is both preliminary and comprehensive. This taxonomy not only encompasses all dark patterns cited in current academic research and regulatory reports, but also increases the two dimensions of severity and impact scenarios to understand the potential impact of dark patterns on users and the different application scenarios they affect.

3.4.2 Conducting the Expert Survey. The primary aim of this online survey is to gather feedback from industry professionals, with a particular focus on designers and UI-related developers. The survey addresses multiple focal points: a). Assess the rationality, completeness, understandability, and usefulness of the dark pattern taxonomy conducted by our work. b). Assess the effectiveness of the annotations in terms of the degree of impact on users and common scenarios. c). Assess the efficacy of existing tools that detect dark patterns. d). Assess the availability of existing dark pattern datasets. e). Collect information from practitioners on their awareness and strategies for identifying and mitigating dark patterns.

The entire survey construction process consists of the following three parts: The first is the design of the questionnaire. The questionnaire comprises 16 questions. Various aspects of focal points are evaluated using a Likert scale [9], ranging from 1 (low) to 5 (high). A minority of questions use binary options, such as whether participants have actually read the taxonomy document and background materials provided by us. Another binary question assesses whether participants feel adequately informed to provide valid feedback. These are critical factors for the quality of the survey data. Additionally, the questionnaire also includes demographic questions that cover participants' age, tenure in their respective industries, and background information. The second is the distribution and scoring method. Snowball sampling techniques are utilized to reach potential participants [28]. Initial recruitment is targeted at designers and UI-related developers with more than 5 years of industry experience, which is later expanded to encompass general practitioners in related fields. The primary channels of recruitment include social media platforms and industry forums. Participants are required to first read the taxonomy document and background materials provided by us and subsequently rate each addressed question. The third is privacy safeguards. To ensure the confidentiality and anonymity of participants, the survey does not collect personally identifiable information. The data collection process adheres to the principle of data minimization. Responses are securely stored, and only aggregated data is reported. The survey is developed using Wjx.cn, a comprehensive platform for survey design, data collection, and analysis. For the actual setting of the questionnaire question, please refer to our website[7]. The survey started on September 1, 2023, and ended on September 25, 2023. The survey collected feedback from 176 respondents.

3.4.3 Updating the Taxonomy. Upon receiving feedback from participants, we will engage in a systematic process to update our existing taxonomy. Initially, we will analyze all the responses collected, meticulously reviewing any contentious points or suggested modifications. Then more interviews with industry experts will be conducted to determine whether the proposed changes should be implemented. Second, any alterations to the taxonomy will be rigorously documented, with annotations explaining the rationale behind each modification. This framework ensures that the taxonomy remains dynamic and evolves in accordance with expert opinions and industry needs.

Furthermore, to account for the dynamic nature of dark patterns, especially as user interfaces continue to evolve and new types of dark patterns emerge, it is imperative to incorporate a robust maintenance mechanism within DPAF. This mechanism would involve periodic reviews and updates to the taxonomy, facilitated by automated scraping algorithms and expert consultations. In doing so, DPAF remains an adaptive tool that not only captures the current state of dark patterns, but also anticipates and adapts to future developments.

3.5 Analysis of Detection Tools

We conducted an in-depth analysis of the 76 articles we collated (refer to Section 3.3) with the aim of unearthing all automated tools adept at identifying dark patterns ③. To streamline our selection process and ensure a focused analysis, we devised the following criteria. Only those tools that met all the criteria were chosen as the subjects of our investigation.

- **Criterion #1 (Dark Pattern Recognition):** It is imperative for the tools to have a direct bearing on dark pattern identification tasks. Their descriptions should encompass verbiage that resonates with keywords such as "identification," "detection," and "classification."
- **Criterion #2 (Automation):** Recognizing that manual identification has inherent drawbacks like increased time consumption and reduced efficiency, our analysis zeroes in exclusively on tools offering automated capabilities.

Table 1. Final Optimized Taxonomy of Dark Patterns

Categories	Sub-Categories	Dark Pattern Types	Sources	Severity	Common Scenarios [§]	AidUI [48]	UIGuard [20]	DY [52]	AGM [50]	ADD [22]	ContextDP-D (301) [5]	Rico-D (1660) [11]	AGM-D (1818) [2]	LLE-D (1787) [4]
Nagging [†]	-	-	[20, 31, 46]	H3,H4,H5	All	✓	✓				✓(57)	✓(188)		✓
Obstruction	Roach Motel	Immortal Accounts [†]	[15, 19, 20, 31, 46, 50]	H1,H2,H3,H5	All			✓	✓	✓			✓(31)	✓
		Dead End [†]	[11]	H1,H2,H4	S				✓	✓				
	Creating Barriers	Forced Grace Period [*]	[51]	H1,H4	All									
		Price Comparison Prevention [†]	[19, 20, 31, 46]	H1,H2,H5,H6	E			✓						✓
	Adding Steps	Intermediate Currency [†]	[20, 31, 46]	H1,H2,H5	S/ES		✓							✓
Sneaking	Bait And Switch	Privacy Maze [†]	[11]	H1,H3	S									
		Labyrinthine Navigation [*]	[51]	H1,H4	S									
		Disguised Ad	[19, 20, 31, 46]	H4,H6	All	✓	✓	✓	✓	✓	✓(21)	✓(46)		✓
	Hiding Information	Sneak Into Basket [†]	[19, 31, 46, 50]	H1,H2,H5	E			✓	✓				✓(21)	✓
		Drip Pricing, Hidden Costs, Or Partitioned Pricing [†]	[19, 20, 31, 46, 50]	H2,H5,H6	E			✓	✓				✓(5)	
(De)Contextualizing Cues		Reference Pricing [†]	[10, 55]	H2,H5,H6	E									
		Conflicting Information	[11, 30]	H4,H6	S									
		Information Without Context	[11]	H4	S									
	Interface Interference	Manipulating Visual Choice Architecture	[20, 31, 46]	H2	All		✓		✓			✓(273)	✓(25)	✓
		Visual Prominence Bundling	[11]	H1,H2	All									
Forced Action	Hard To Close *	Pressured Selling [†]	[10]	H1	E				✓				✓(67)	
		Persuasive Language [†]	[21, 50]	H2,H4,H6	E									
		Small or Moving Close Button [*]	[51]	H1,H2,H4	E/S									
		Bad Defaults/Pre-Selection [†]	[20]	H4	All		✓					✓(684)		
		Emotional Or Sensory Manipulation	[15, 20, 31, 46]	H1,H2,H3,H6	S	✓	✓				✓(111)	✓(232)		✓
	Trick Questions [†]	Cuteness [†]	[44, 50]	H2,H5	E									
		Positive Or Negative Framing	[20, 50]	H2,H3,H4	All		✓							✓
		Fear Of Missing Out [†]	[20, 50]	H2,H3,H4	All									
		Choice Overload	[19, 20, 31, 46, 50]	H2,H4,H6	E			✓	✓	✓			✓(9)	✓
		Hidden Information [†]	[10, 11]	H4	S									
	Language Inaccessibility	Wrong Language [†]	[31, 46]	H3,H5,H6	All									✓
		Complex Language	[11]	H3,H4	All									
		Feedforward Ambiguity	[10]	H4	All									
		Plain Evil [*]	[11, 30]	H4	All									
		Engaging Strategy [*]	[51]	H1,H4	S									
Social Engineering	Forced Continuity [†]	Addictive Design [*]	[51]	H1,H4	S									
		Infinite Scrolling [*]	[51]	H1,H4	S									
		Pull To Refresh [*]	[51]	H1,H4	S									
		Reduced Friction [*]	[51]	H4	S									
		Customisation [*]	[51]	H1,H2,H5	E			✓	✓			✓(1)		
	Forced Registration Or Disclosure	Privacy Zuckering [†]	[19, 20, 31, 46, 50]	H1,H2,H5	E			✓	✓				✓(6)	✓
		Friend Spam [†]	[15, 46, 50]	H1,H5	E/S				✓			✓(117)		✓
		Address Book Leeching [†]	[19, 20, 31, 46]	H1,H3	E/S			✓						
		Social Pyramid [†]	[15, 46]	H1,H3	E/S									
		Granting and Interaction [†]	[20, 31, 46]	H1,H3	All		✓					✓(16)		✓
	Gamification	Pay-To-Play [†]	[51]	H1	S									
		Grinding	[21, 31, 46]	H1,H2	ES	✓	✓				✓(11)			
		Playing By Appointment [*]	[21, 31, 46, 61]	H1,H4	ES	✓	✓							
		Countdown On Ads [*]	[61]	H1	ES									
		Watch Ads To Unlock Features Or Get Rewards [*]	[20]	H4	All			✓				✓(4)		✓
Pre-Delivered Content [†]	Automatic Execution [*]	Pay To Avoid [*]	[20]	H4	All			✓				✓(2)		✓
		Automating The User Away [*]	[30]	H1,H4	ES			✓				✓(97)		✓
		Automatic Accept Third Party Term [*]	[51]	H1	S									
		Attention Capture	[21]	H1,H5	All	✓		✓			✓(13)			
		Auto-Play	[46, 50]	H2,H6	E	✓			✓		✓(5)		✓(47)	
	Social Proof	High Demand [†]	[46, 50]	H2,H6	E									
		Low Stock [†]	[46, 50]	H2,H6	E	✓			✓		✓(19)		✓(632)	
		Endorsement And Testimonials	[46, 50]	H2,H6	E				✓				✓(12)	
		Parasocial Pressure [†]	[21, 46, 50]	H2,H6	E									
		Activity Messages [†]	[46, 50]	H2,H6	E	✓			✓		✓(10)		✓(313)	
	Urgency	Countdown Timer	[46, 50]	H2,H6	E	✓			✓		✓(28)		✓(393)	
		Limited Time Message [†]	[46, 50]	H2,H6	E	✓			✓		✓(26)		✓(88)	
		Confirmshaming [†]	[19, 46, 50]	H2	E			✓	✓				✓(169)	
		Encouraging Anti-Social Behavior [*]	[61]	H4	ES									
		Psychological Tricks [*]	[61]	H4,H6	ES									
	Shades Of Grey [*]	Games For Other Purposes [*]	[61]	H4,H6	ES									
		Pre-Delivered Content [†]	-	H2	ES									

§: "Common scenarios" refers to the application scenarios in which the dark mode is present. It encompasses four distinct categories: Social Platforms (S), E-commerce (E), Entertainment Software (ES), and General or Universal (All). Dark patterns identified with a '†' symbol signify actions or features that result in financial losses and privacy harms for users, highlighting the necessity to prioritize their detection in the realm of dark pattern recognition. Dark patterns labeled with a '*' denote newly introduced dark patterns.

Table 2. Tools List for Dark Patterns Detection

Tools	Technology	Platform	Automaticity	Availability
AidUI [48]	Visual Cue Detection based on deep learning; Spatial Analysis; Text Analysis; Color Analysis.	Web & Mobile	Automatic	Yes [6]
Ulguard [20]	UI extraction based on Faster-RCNN; Text Analysis; Icon Semantic Understanding; UI Color Extraction; Knowledge-Driven checker	Mobile	Automatic	No
DY [52]	Hierarchical and k-means cluster analysis algorithms	Web	Automatic	No
AGM [50]	Corpus creation; Crawler; Hierarchical Clustering using HDBSCAN; Manual Examination	Web	Semi-Automatic	Yes [3]
ADD [22]	Web crawling and web scraping techniques; Natural language processing	Web	Automatic	No
SSS [58]	Text extraction and analysis	Web	Automatic	No

- **Criterion #3 (Functional Description):** Clarity is crucial. As such, each tool must be supported by comprehensive and clear documentation, to ensure a seamless evaluation of its features and functionalities.
- **Criterion #4 (Types of Dark Patterns Detected):** The documentation of each tool should elucidate the specific dark patterns that it can detect.

In light of criteria #1 through #3, six tools emerged as potent candidates for dark pattern recognition. Table 2 provides a detailed overview of these tools. The first column lists the tool's name and source. In instances where tools lacked a designated name, we adopted an abbreviation strategy, leveraging the initial letters of the first three authors of the corresponding paper. The next column outlines the primary technologies employed by each tool. This is followed by a column specifying the tool's target platform, either web or mobile. Lastly, columns indicate the tool's automation capability and its public availability. Furthermore, RQ2 underscores the importance of dissecting the prowess of various tools in the realm of dark pattern identification. However, as Table 2 reveals, most of these tools are not open source. Therefore, for such tools, it is crucial that their documentation transparently enumerates the dark patterns that they are proficient in detecting. Adherence to criterion #4 resulted in the exclusion of SSS [58] from our list. Consequently, our finalized list for in-depth study comprises five tools: AidUI[48], Ulguard[20], DY[52], AGM [50], and ADD[22].

In our evaluation of selected dark pattern detection tools, we employed a manual marking strategy to categorize their claimed or actual detection capabilities within the taxonomy we developed. Initially, we conducted a comprehensive review of the academic literature, documentation, and code of each tool, open source or commercial, to meticulously compile a list of dark pattern types that each tool purportedly or effectively recognizes. Following this process, we systematically annotated these capabilities within our constructed dark pattern taxonomy. During the practical evaluation phase, out of the six tools evaluated, only AidUI[48] was executable, with the others either lacking accessible source or executable code or having non-functional source code. For AidUI, we conducted an in-depth analysis of its detection capabilities after execution. For tools that could not be executed, we relied on an extensive review of their source code and documentation to evaluate their claimed detection abilities.

It is worth noting that if a certain dark pattern a tool claims to identify has been merged with other similar dark patterns in our taxonomy, the newly amalgamated pattern within our taxonomy is annotated as recognizable.

3.6 Analysis of Datasets

To explore whether current public datasets suffice for studies related to dark patterns, we embarked on an evaluation of the comprehensiveness of these datasets in terms of dark patterns④. To achieve this, we established precise criteria for the selection of the dataset, ensuring alignment with the intricacies of dark pattern detection.

Table 3. Datasets List for Dark Patterns

Dataset	C/J	Publish Year	Platform	# Instances	#Patterns coverage	Availability
ContextDP-D [48]	ICSE	2023	Web & Mobile	301	10	Yes [5]
Rico'-D [20]	HCI	2023	Mobile	1,660	12	Yes [1]
AGM-D [50]	HCI	2019	Web	1,818	15	Yes [2]
LLE-D [24]	CHI	2020	Mobile	1787	16	Permission required [4]

Initially, we delved into our repository of 76 dark pattern literature, aiming to pinpoint all papers that describe or allude to datasets containing dark pattern instances. Our dataset acquisition was steered by two fundamental criteria:

- **Criterion #1 (Presence of Dark Pattern Instances):** The dataset should explicitly contain screenshots or videos of web pages or mobile apps with dark patterns. This provides researchers with a solid foundation from which to extract and comprehend the features of dark patterns.
- **Criterion #2 (Open Accessibility):** The dataset must be publicly disclosed by its authors and readily available without constraints.

Employing these criteria, our literature review yielded 4 articles that touched on datasets capturing dark pattern instances. Among these, 3 were primarily oriented toward the exploration of dark pattern detection tools, while one dedicated its focus to charting the prevalence of dark patterns. Table 3 provides a detailed overview of the primary information for these four datasets. The first column describes the name and source of the datasets. Notably, in instances where the datasets lacked a distinct designation by their authors, we instituted a uniform naming convention: using the initials of the first three authors, complemented by the "-D" suffix. This is followed by the conference or journal of the associated paper and its publication year. The fourth column specifies the target platform, either web or mobile. The fifth and sixth columns depict the scale of the dataset (number of instances it encompasses) and the count of dark pattern types it covers, respectively. The last column highlights the availability, complemented by a direct link if applicable. As indicated in the table, except for LLE-D [24], all other datasets are directly accessible.

For the four datasets selected, we enumerated the types of dark patterns present within each and documented their respective instance counts. Within our dark pattern taxonomy, we denoted the spectrum of dark patterns encompassed by each dataset. Consistent with our prior approach to marking tool capability, if a covered dark pattern has been merged with another dark pattern, we would denote the merged dark pattern as present in this dataset.

4 RESULTS AND INSIGHTS

This section introduces the dark pattern taxonomy constructed based on our proposed **DPAF** framework, as well as the annotations regarding tool capabilities and data availability made on the taxonomy, to address the three research questions.

4.1 Taxonomy

For RQ1, after following the steps outlined in Section 3.4, from the 76 scientific literature, we have constructed the most comprehensive and standardized taxonomy of dark patterns to date, as shown in Table 1.

In general, this taxonomy encompasses 64 *Dark Patterns Types* of dark patterns (in the third column). These types belong to 6 categories (in the first column) and 31 sub-categories (in the second column). Compared to the most recent taxonomy [32], our taxonomy has added 7 new *sub-categories* and 20 new *Dark Patterns Types*. These newly added patterns are annotated with asterisks

Table 4. Feedback Results of Survey

# Feedback	# Valid feedback	Rationality	Completeness	Understandability	Usefulness	Severity Annotation	Scenarios Annotation
176	173	100%	92%	87%	92%	100%	100%

in Table 1. The 'Severity' column indicates the types of impact each dark pattern has on users. Among the identified dark patterns, 49 of 64 types, which account for 77%, exhibit multifaceted impacts on users, as evidenced by their association with multiple labels. Among all the H1-H6 impact labels, we find that H2 and H3 are the most severe, as they cause financial loss and privacy leakage, respectively. In the table, 33 dark patterns are labeled with these two severe types. If priority ranking is needed for dark pattern identification tasks, those labeled H2 and H3 should be identified first. The 'Common Scenarios' column specifies the potential settings where each type of dark pattern may appear. Fifteen types of dark patterns are prevalent in all three business scenarios: *Social Platforms (S)*, *E-Commerce (E)*, and *Entertainment Software (ES)*, demonstrating their ubiquity. Furthermore, 17 types are unique to *Social Platforms*, 17 to *E-Commerce*, and 9 to *Enterprise Software*, with 6 types appearing in both *E* and *S* settings. Analyzing the likely scenarios for each dark pattern type helps to provide information support to stakeholders such as researchers, designers, and policymakers.

These findings have been widely supported by industry experts. Specifically, as shown in Table 4, an analysis of the feedback from all 173 valid participants reveals that more than 87% believe our dark patterns taxonomy is essentially rationality, completeness, understandability, and usefulness, rating it 4 or 5 points. The general consensus is that our annotations on *Severity* and *Common Scenarios* are reasonable. Furthermore, we received suggestions from 10 participants who recommended including screenshots for each dark pattern type for a better understanding. This was the main reason for the slightly lower approval rate with respect to comprehensibility. Due to the considerable manual annotation work involved, we plan to introduce an interactive website in future research to facilitate a more intuitive understanding of dark patterns[7]. Furthermore, through the survey, we found that 61% of the participants were aware of the existence of dark patterns in the apps they developed, 23% were not aware, and 15% were uncertain. In terms of avoiding dark patterns, 30% reported that their organizations had adopted strategies to evade them, 23% said that no such measures were in place, and the remaining 46% were unaware. This indicates a general lack of awareness and proactive measures against dark patterns in the industry. For more information, please refer to our website.

Answer to RQ1: Through a review of 76 scientific papers, we identified and generalized 64 unique types of dark patterns. We further organized and annotated these patterns according to their categories, descriptions, impacts on users, and common scenarios. This taxonomy was validated by industry participants, with more than 87% of responses deeming it rationality, completeness, understandability and usefulness.

4.2 Capability of Detection Tools

For RQ2, our objective is to evaluate the capabilities of various dark pattern detection tools in alignment with our comprehensive taxonomy. Following the procedures introduced in Section 3.5, we selected a total of five different dark pattern detection tools, including AidUI[48], UIGuard[20], DY[52], AGM[50], and ADD[22]. As illustrated in Table 2, among the five tools, three are designed to detect dark patterns in web pages, one is designed for mobile apps, and the other is for both platforms. These tools use advanced techniques to identify dark patterns, such as machine learning and natural language processing. Except for AGM, which automates most of its processes but still

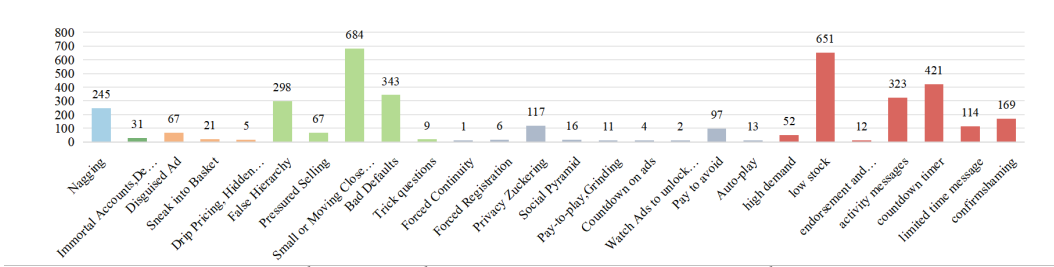


Fig. 3. Comprehensive Dark Pattern Instances Count Across Three Datasets

requires manual intervention, all other tools claim to be fully automated. In addition, among these tools, AidUI and AGM are open source, while other tools are currently not available.

To further investigate the efficacy of these tools, Table 1 delves into the detection capabilities of these tools in relation to our comprehensive dark pattern taxonomy. The 7 to 11 columns of Table 1 correspond to each tool, marked with ✓ if the tool can identify the respective dark pattern. As illustrated in the table, these five tools collectively detect 32 distinct dark patterns, achieving a coverage rate of 50% ($=32/64$). Most detectable dark patterns can be identified using only one or two tools. Specifically, 11 of these dark patterns are detected solely by one tool, while 20 are identified by 2 or 3 tools. Remarkably, only one dark pattern (i.e., "Disguised Ad") can be recognized by four of the tools. Intriguingly, none of these 32 dark patterns is universally detected across all tools. Considering the capabilities of individual tools, they can identify between 5 to 15 types of dark patterns. Specifically, UIGuard and AGM can detect the most, identifying up to 15 dark patterns, while ADD detects the fewest, pinpointing only 5 dark patterns. It suggests that these tools, employing varied technical approaches and focusing on different research aspects, demonstrate significant discrepancies in the types of dark patterns that they can identify. Moreover, in relation to the 64 dark patterns we have cataloged, the current tools demonstrate a relatively low recognition rate. The highest recognition rate is only 23.4%, that is, 15 of 64. This underscores the inadequacy of the detection capabilities of existing tools.

Additionally, there are 32 dark patterns, comprising 50% of the total, that remain undetected by any of the tools. The inability to detect these patterns could be attributed to various reasons: certain patterns might be inherently intricate, heightening detection challenges; some might manifest in diverse manners, complicating the formulation of clear detection rules; or certain patterns may seldom surface in real-world scenarios, leading to an insufficiency of representative samples, which in turn hampers a comprehensive understanding and identification. These observations highlight that the current dark pattern detection tools have substantial limitations, reflecting a lack of comprehensiveness and robustness.

The results demonstrate a significant gap between the breadth of dark patterns identified in our taxonomy and those currently detectable by existing tools. This underscores the need for more advanced and comprehensive tools to protect users from a wider array of dark patterns.

Answer to RQ2: From the five tools selected, they collectively identify 32 out of the 64 dark patterns from our taxonomy, achieving a 50% coverage. However, 32 dark patterns (50% of the total) remain undetected by all the tools.

4.3 Availability of Datasets

For RQ3, through an in-depth analysis of the selected datasets, our objective is to examine their coverage in relation to the taxonomy we have established, allowing us to assess the comprehensiveness and integrity of these datasets.

As described in section 3.6, we selected a total of four datasets containing dark pattern instances, including ContextDP-D [48], Rico'-D [20], AGM-D [50] and LLE-D [24]. Table 3 reveals a collective count of 5,566 instances in these datasets. More specifically, Rico'-D and LLE-D contain 1,660 and 1,787 dark pattern instances from mobile apps, respectively. AGM-D encompasses 1,818 instances exclusively from web pages. And ContextDP-D has a smaller count of 301 instances from both platforms.

To delve deeper into the current data coverage of dark patterns, Table 1 elucidates the types of dark patterns encompassed by each dataset. The final four columns correspond to each of the datasets. When a particular dark pattern instance is present within a dataset, we mark the respective cell as "covered" with a checkmark and denote the specific instance count of that dark pattern. After merging these datasets, Figure 3 provides a statistical count of the instances for each covered dark pattern, represented in the form of a bar chart. It is worth noting that while the LLE-D dataset is claimed publicly available, it requires permission, hindering direct access and retrieval of the data. Consequently, in our evaluation of the comprehensiveness of this dataset, we primarily rely on descriptions provided in the associated literature. Only when it explicitly mentions the inclusion of a specific dark pattern instance within its dataset do we mark it as covered in our table. Given our inability to access this dataset, we remain uninformed about the precise count of certain dark pattern instances. Hence, in subsequent result descriptions, statistics involving instance counts will exclude this particular dataset.

From Table 1, it is discernible that these four datasets collectively cover only 32 types of dark patterns, resulting in an overall coverage rate of 50% (that is, 32/64). When considering each dataset individually, the range of covered dark patterns narrows further, capturing merely 10 to 16 distinct types. Furthermore, as shown in Figure 3, there is a significant disparity in the number of instances for each type of dark pattern within the datasets. In particular, the instance counts for "Low Stock" and "Small or Moving Close Button" stand out, both exceeding 600 instances. This suggests that these dark patterns might be more prevalent in everyday applications, making their instances more readily collectible. In contrast, 6 dark patterns exhibit substantially fewer instances, with types like "Tick questions", "Forced Continuity" and "Countdown on Ads" having counts of less than ten. This pronounced data imbalance could potentially jeopardize the performance of machine learning-based dark pattern prediction and classification, introducing model biases and thereby compromising model accuracy and robustness.

Furthermore, the absence of 32 types of dark patterns (accounting for 50%) in the datasets implies that machine learning models may fail to acquire the features of these patterns, rendering them ineffective in recognizing and classifying such instances. The possible reasons for the absence of these dark patterns in the datasets can be delineated as follows: (1) Ambiguity in Definitions and Absence of Precise Criteria: The ambiguity surrounding the definitions of some dark patterns often impedes researchers from conclusively identifying them on screencasts of mobile applications or web pages. For instance, while "complex language" is described as "Decision-relevant information can be deliberately or unintentionally made difficult to understand by using complex language" [10, 56], there is no concrete benchmark that establishes what qualifies as complex language. This lack of clarity might lead researchers astray when trying to pinpoint such patterns. (2) Dependence on Contextual and Historical User Interaction: Some dark patterns are not confined to a singular interface and manifest themselves based on user-interaction history or a broader context. As such, they elude immediate detection on standalone interfaces, making their data collection challenging. An illustrative example is "Address Book Leeching", where services conduct database searches upon users importing their contact lists. Although these services may store these contact details for ulterior reasons, not initially disclosed to users, malicious intent may surface only at a later interaction juncture [15]. The current method of collecting dark pattern instance data mostly

relies on manual identification and labeling of screenshots or videos of mobile applications or web pages, which lacks the capability to identify such nuanced patterns based on a single interface. Coincidentally, both detection tools and datasets cover 50% of the dark pattern taxonomy but exhibit slight variations in the types of dark pattern they cover. Both cover 31 types, with differences in Friend Spam (detectable but not covered by the datasets) and Hidden Information (covered by the datasets but undetectable by the tools).

The findings from our dataset analysis carry significant implications for the domain of dark pattern detection using machine learning techniques. The evident data imbalance, with certain patterns being over-represented and others under-represented or even absent. Models trained on such datasets might exhibit heightened sensitivity to more frequent patterns and diminished capability to recognize rarer ones. Additionally, the complete absence of 32 dark pattern types underscores a glaring gap in current datasets, limiting the model's capacity to detect a wide spectrum of dark patterns. Hence, for reliable and holistic dark pattern detection, there is a pressing need for more balanced and exhaustive datasets, encompassing the diversity of dark patterns encountered in real-world scenarios.

Answer to RQ3: The four datasets collectively contain 5,566 instances, and only cover 32 distinct dark patterns, yielding a total coverage rate of 50% (i.e., 32 out of 64 types). The omission of the other 32 dark patterns, which also make up 50% of the total, restricts the ability to recognize these patterns.

5 DISCUSSION

5.1 Implications

Our research in constructing a comprehensive taxonomy for dark patterns has multiple implications for both the academic and industrial communities.

- *Standardization:* The taxonomy provides a standard nomenclature for classifying dark patterns, thus reducing ambiguity and facilitating better communication among researchers and practitioners.
- *Impact Assessment:* By annotating the dimensions of user impact and potential application scenarios for each dark pattern, we offer a nuanced understanding that can guide design decisions and policy interventions.
- *Tool Evaluation:* Our annotation of the detection capabilities of existing tools serves as a benchmark for evaluating and improving dark pattern detection mechanisms.
- *Data-Driven Insights:* With the inclusion of the number of available instances for each type of dark pattern, our taxonomy becomes a data-driven tool that can guide future research and applications.

5.2 Threats to Validity

5.2.1 Internal Threats. The manual labeling process for identifying the dimensions of user impact and application scenarios is prone to subjectivity and potential inaccuracies. Although we mitigated this risk through expert consultations, the element of human error cannot be completely eliminated. The study relies on industry surveys to validate the taxonomy. The quality of this validation is highly dependent on the respondent's understanding and interpretation of the taxonomy, thereby introducing another layer of subjectivity. The methods used to evaluate existing dark pattern detection tools may not cover all the possible configurations and nuances, thus affecting the completeness of our evaluation.

5.2.2 External Threats. First, our validation is based on surveys from industry professionals. However, the samples may not be fully representative of the diverse practitioners in this field. Second, the study focuses primarily on industry professionals who are more likely to be familiar with western design paradigms. The taxonomy and its implications might not be universally applicable in different cultural and geographical contexts. Third, we evaluated existing dark pattern detection tools based on their capabilities at the time of our study. These tools themselves are subject to updates and improvements that might affect the relevance of our assessment in the future.

5.3 Challenges

The research landscape of dark patterns is complex due to advanced algorithms, data challenges, and tool usability. Our study provides a foundational framework but faces several avenues for future research and challenges. First, the quality and quantity of data are essential for the robustness of our taxonomy. Despite efforts to ensure accuracy through expert consultations, the manual labeling process still bears the risk of error and subjectivity. Second, the development of open-source tools for detecting dark patterns is crucial but comes with evaluation challenges. Current methodologies may not fully capture the diversity and subtleties of dark patterns, risking bias, and undermining generalizability. Third, longitudinal studies[12], while valuable, are complicated by extended time frames, analytical complexity, and the dynamic nature of digital interfaces. These factors could compromise the reliability and generalizability of the findings. Lastly, ethical considerations are paramount. With increasing regulatory focus, agile ethical guidelines are needed to accompany our taxonomy, which requires a dynamic approach to its development.

In summary, our study is a stepping stone in the field of dark patterns, and ongoing multidisciplinary efforts are needed to refine methodologies and tools to adapt to the ever-evolving nature of dark patterns.

6 RELATED WORK

In this section, we describe previous research related to dark patterns, with a focus on their classification and detection. We begin by summarizing the work that classifies dark patterns through the proposal of taxonomies and that suggests techniques for detecting these patterns within GUIs. Subsequent subsections elaborate on the classification and detection aspects.

6.1 Classification of Dark Patterns

Numerous empirical studies have explored the concept of dark patterns on various digital platforms, including websites, mobile apps [23, 31, 50], and diverse domains such as gaming [8], social media platforms [36], e-commerce websites [50], cookie banners [33, 53], and advertisements [27]. The typical approach in these empirical studies involves the collection and examination of user interfaces from desktop or mobile applications. The researchers then consolidate their findings, identifying and categorizing the types of dark patterns embedded in these interfaces.

This journey began in 2010 when Harry Brignull [17] established the concept of dark patterns, defining them as manipulative techniques employed by websites and apps to force users into unintended actions, such as making purchases or signing up for services. Brignull also launched a Twitter account [16] for users to report and discuss instances of dark patterns they encountered in their daily digital interactions. Subsequently, Gray et al. [31] conducted manual data collection from popular online platforms over a two-month period. They expanded Brignull's concept by classifying dark patterns into five strategic categories: nagging, obstruction, sneaking, etc. Building on these taxonomies, Mathur et al. [50] executed a large-scale empirical study, focusing on around 11,000 shopping websites. Their approach involved simulating user shopping behaviors and clustering segments of the collected websites. In contrast to Mathur et al.'s concentration on e-commerce

platforms, Di Geronimo et al. [23] examined the presence of dark pattern elements in 240 widely used Android applications. They achieved this by recording 10-minute usage videos for each application and adopted taxonomies derived from Gray et al. [32]. Their work identified 16 new dark pattern types comprising 31 cases.

The research most relevant to our study is conducted by Gray et al. [32], who developed a comparatively comprehensive taxonomy of dark patterns. Unlike their work, in this paper, we synthesize all previous taxonomies through a systematic literature analysis and, building on the taxonomy established by Gray et al., introduce 20 important and previously unrecorded types. Furthermore, based on the taxonomy we constructed, we meticulously mark and analyze the recognition capabilities of existing detection tools and the coverage rate of datasets.

6.2 Detection of Dark Patterns

Various strategies have been employed to detect dark patterns, ranging from manual exploration to semi-automated clustering methods and basic text-based classification techniques.

Detection of dark patterns was initially based on manual exploration, conducted by domain experts [17, 23, 31, 36, 38] or ordinary users [16]. To address the limitations of manual exploration, Mathur et al. [50] introduced semi-automated techniques. These methods simulate user behavior to uncover dark patterns within shopping websites. They employed clustering techniques to group related UI patterns, making the detection process more efficient. Based on their findings and dataset, some researchers [60] explored classification techniques to detect dark patterns in real-world scenarios. However, these techniques are text-based and limited in scope, possibly resulting in numerous false positives. Additionally, they rely on labeled training data for dark patterns. Recently, Daniel [38] developed DarkDialogs, a system designed to autonomously extract consent dialogs from websites and identify the existence of 10 dark patterns. After that, more and more automated dark pattern detection tools were proposed. Andrea et al. [22] combined natural language processing techniques with web crawling and scraping techniques to design and develop a framework to automatically detect potential instances of web-based dark patterns. For the mobile app, Hrushikesava et al. [58] designed an application through which any identified dark pattern advertisement can be viewed and warned through a dialog box. Dmitry et al. [52] proposed to implement dark pattern detection on user interface sites using a cluster analysis algorithm that uses both hierarchical and k-means methods to cluster many dark patterns in the application interface. In the latest dark pattern detection tools, image processing has emerged as a key technique to improve the capability of dark pattern detection tools. Mansur et al. [48] proposed the AidUI tool by combining techniques such as visual processing and text analysis, which demonstrated excellent performance for dark pattern detection. Chen et al. [20] proposed UIGuard by extracting the UI interface and then combining the text analysis of the UI interface along with icon semantic understanding and UI color analysis, obtaining good results. Based on previous work, our paper assesses the capabilities and limitations of existing tools for detecting dark patterns in the taxonomy we built. We also examine various benchmark datasets used to detect dark patterns.

7 CONCLUSION

In summary, this study represents a significant milestone in the rapidly evolving field of dark pattern research. Through a meticulous systematic review, we have identified 76 seminal papers, allowing us to construct the most comprehensive standardized taxonomy of dark patterns to date. This taxonomy not only includes types and descriptions of each dark pattern, their categorizations, but also annotates the impact on users and potential application scenarios. Furthermore, we have marked the capabilities of existing dark pattern detection tools and the availability of current data

within this taxonomy. Our taxonomy serves as both a standard nomenclature for classifying dark patterns and a practical tool for researchers, practitioners, and policymakers alike.

The road ahead is fraught with challenges that require a multidisciplinary approach and ongoing refinement of methods and tools. The landscape of dark patterns is in constant flux, influenced by rapid technological advancements and shifting societal norms. As such, future research must focus on developing more robust methodologies, including open-source tools and ethical guidelines, that can adapt to this ever-changing landscape.

ACKNOWLEDGMENTS

We thank the reviewers for their insightful comments and suggestions. This work was partially funded by the National Natural Science Foundation of China (61972359, 62132014), Zhejiang Provincial Key Research and Development Program of China (2022C01045), Zhejiang Provincial Natural Science Foundation of China (LQ23F020020), the Education Department of Hunan Province (21B0313), the Natural Science Foundation of SZTU Top Talent (GDRC202132), the SZTU-Enterprise Cooperation Project (20221061030002, 20221064010094), the Inner Mongolia Key Project (2021ZD0044), and the SZTU Project (20224027010006).

REFERENCES

- [1] 2017. *Dataset:Rico*. Retrieved Sept. 28, 2023 from <https://www.interactionmining.org/>
- [2] 2019. *Dataset:AGM*. Retrieved Sept. 28, 2023 from <https://darkpatterns.cs.princeton.edu/data/>
- [3] 2019. *Tool:AGM*. Retrieved Sept. 28, 2023 from <https://github.com/aruneshmathur/dark-patterns>
- [4] 2020. *Dataset:LLE*. Retrieved Sept. 28, 2023 from <https://zenodo.org/deposit/3601501>
- [5] 2023. *Dataset:ContextDP*. Retrieved Sept. 28, 2023 from <https://github.com/SageSELab/AidUI/>
- [6] 2023. *Tool:AidUI*. Retrieved Sept. 28, 2023 from <https://github.com/SageSELab/AidUI/>
- [7] 2023. *websit:darkpatternsok*. Retrieved Sept. 28, 2023 from <https://sites.google.com/view/darkpattern>
- [8] Jacob Aagaard, Miria Emma Clausen Knudsen, Per Bækgaard, and Kevin Doherty. 2022. A Game of Dark Patterns: Designing Healthy, Highly-Engaging Mobile Games. , 438:1–438:8 pages. <https://doi.org/10.1145/3491101.3519837>
- [9] Gerald Albaum. 1997. The Likert scale revisited. *Market Research Society. Journal*. 39, 2 (1997), 1–21. <https://doi.org/10.1177/147078539703900202>
- [10] Competition Market Authorities. 2022. Evidence Review of Online Choice Architecture and Consumer and Competition Harm.
- [11] European Data Protection Board. 2022. *Guidelines 3/2022 on Dark patterns in social media platform interfaces: How to recognise and avoid them*. https://edpb.europa.eu/system/files/2022-03/edpb_03-2022_guidelines_on_dark_patterns_in_social_media_platform_interfaces_en.pdf Technical Report Version 1.0.
- [12] Niall Bolger, Gertraud Stadler, and Jean-Philippe Laurenceau. 2012. Power analysis for intensive longitudinal studies. (2012).
- [13] Sebastian Boring, Saul Greenberg, Jo Vermeulen, Jakub Dostal, and Nicolai Marquardt. 2014. The dark patterns of proxemic sensing. *Computer* 47, 8 (2014), 56–60. <https://doi.org/10.1109/MC.2014.223>
- [14] Christoph Bösch, Benjamin Erb, Frank Kargl, Henning Kopp, and Stefan Pfattheicher. 2016. Tales from the Dark Side: Privacy Dark Strategies and Privacy Dark Patterns. *Proceedings on Privacy Enhancing Technologies* 2016, 4 (2016), 237–254. <https://doi.org/10.1515/popets-2016-0038>
- [15] Christoph Bösch, Benjamin Erb, Frank Kargl, Henning Kopp, and Stefan Pfattheicher. 2016. Tales from the dark side: privacy dark strategies and privacy dark patterns. *Proc. Priv. Enhancing Technol.* 2016, 4 (2016), 237–254.
- [16] Harry Brignull. 2010. Twitter: Deceptive Design@darkpatterns. *Twitter* (2010).
- [17] Harry Brignull. 2010. Types of deceptive design. (2010). <https://www.deceptive.design/types>
- [18] Harry Brignull. 2011. Dark Patterns: Deception vs. Honesty in UI Design. *Interaction Design, Usability* 338 (2011), 2–4.
- [19] Harry Brignull, Marc Miquel, Jeremy Rosenberg, and James Offer. 2015. Dark patterns-user interfaces designed to trick people. In *Proceedings of the Poster Presentation, Australian Psychological Society Congress, Sydney, NSW, Australia*. 21–23.
- [20] Jieshan Chen, Jiamou Sun, Sidong Feng, Zhenchang Xing, Qinghua Lu, Xiwei Xu, and Chunyang Chen. 2023. Unveiling the Tricks: Automated Detection of Dark Patterns in Mobile Applications. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*. 1–20.
- [21] Federal Trade Commission. 2022. *Bringing Dark Patterns to Light Staff Report*. https://www.ftc.gov/system/files/ftc_gov/pdf/P214800%20Dark%20Patterns%20Report%209.14.2022%20-%20FINAL.pdf Technical Report Version.

- [22] Andrea Curley, Dympna O'Sullivan, Damian Gordon, Brendan Tierney, and Ioannis Stavrakakis. 2021. The design of a framework for the detection of web-based dark patterns. (2021).
- [23] Linda Di Geronimo, Larissa Braz, Enrico Fregnan, Fabio Palomba, and Alberto Bacchelli. 2020. UI Dark Patterns and Where to Find Them: A Study on Mobile Applications and User Perception. , 14 pages. <https://doi.org/10.1145/3313831.3376600>
- [24] Linda Di Geronimo, Larissa Braz, Enrico Fregnan, Fabio Palomba, and Alberto Bacchelli. 2020. UI dark patterns and where to find them: a study on mobile applications and user perception. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–14.
- [25] Gregory M. Dickinson. 2023. Privately Policing Dark Patterns. <https://doi.org/10.2139/ssrn.4389572>
- [26] Madison Fansher, Shruthi Sai Chivukula, and Colin M Gray. 2018. # Darkpatterns: UX practitioner conversations about ethical design. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–6. <https://doi.org/10.1145/3170427.3188553>
- [27] Liza Gak, Seyi Olojo, and Niloufar Salehi. 2022. The distressing ads that persist: Uncovering the harms of targeted weight-loss ads among users with histories of disordered eating. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–23. <https://doi.org/10.1145/3555102>
- [28] Leo A Goodman. 1961. Snowball sampling. *The annals of mathematical statistics* (1961), 148–170.
- [29] Colin M Gray, Lorena Sánchez Chamorro, Ike Obi, and Ja-Nae Duane. 2023. Mapping the Landscape of Dark Patterns Scholarship: A Systematic Literature Review. , 188–193 pages. <https://doi.org/10.1145/3563703.3596635>
- [30] Colin M Gray, Shruthi Sai Chivukula, and Ahreum Lee. 2020. What Kind of Work Do " Asshole Designers" Create? Describing Properties of Ethical Concern on Reddit. In *Proceedings of the 2020 acm designing interactive systems conference*. 61–73.
- [31] Colin M Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L Toombs. 2018. The dark (patterns) side of UX design. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. <https://doi.org/10.1145/3173574.3174108>
- [32] Colin M Gray, Cristiana Santos, and Nataliia Bielova. 2023. Towards a Preliminary Ontology of Dark Patterns Knowledge. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–9. <https://doi.org/10.1145/3544549.3585676>
- [33] Colin M Gray, Cristiana Santos, Nataliia Bielova, Michael Toth, and Damian Clifford. 2021. Dark patterns and the legal requirements of consent banners: An interaction criticism perspective. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–18. <https://doi.org/10.1145/3411764.3445779>
- [34] Colin M. Gray, Cristiana Teixeira Santos, Nicole Tong, Thomas Mildner, Arianna Rossi, Johanna T. Gunawan, and Caroline Sindors. 2023. Dark Patterns and the Emerging Threats of Deceptive Design Practices. (2023), 510:1–510:4. <https://doi.org/10.1145/3544549.3583173>
- [35] Johanna Gunawan, Amogh Pradeep, David Choffnes, Woodrow Hartzog, and Christo Wilson. 2021. A Comparative Study of Dark Patterns Across Web and Mobile Modalities. *Proc. ACM Hum.-Comput. Interact.* 5, Article 377 (oct 2021), 29 pages.
- [36] Hana Habib, Sarah Pearman, Ellie Young, Ishika Saxena, Robert Zhang, and Lorrie Falth Cranor. 2022. Identifying user needs for advertising controls on Facebook. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW1 (2022), 1–42. <https://doi.org/10.1145/3512906>
- [37] Shun Hidaka, Sota Kobuki, Mizuki Watanabe, and Katie Seaborn. 2023. Linguistic Dead-Ends and Alphabet Soup: Finding Dark Patterns in Japanese Apps. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–13. <https://doi.org/10.1145/3544548.3580942>
- [38] Daniel Kirkman, Kami Vaniea, and Daniel W. Woods. 2023. DarkDialogs: Automated detection of 10 dark patterns on cookie dialogs. , 847–867 pages. <https://doi.org/10.1109/EuroSP57164.2023.00055>
- [39] Barbara Kitchenham, O Pearl Brereton, David Budgen, Mark Turner, John Bailey, and Stephen Linkman. 2009. Systematic literature reviews in software engineering—a systematic literature review. *Information and software technology* 51, 1 (2009), 7–15. <https://doi.org/10.1016/j.infsof.2008.09.009>
- [40] Agnieszka Kitkowska. 2023. The Hows and Whys of Dark Patterns: Categorizations and Privacy. *Human Factors in Privacy Research* (2023), 173–198.
- [41] Tim Kollmer and Andreas Eckhardt. 2023. Dark Patterns: Conceptualization and Future Research Directions. *Business and Information Systems Engineering* 65, 2 (2023), 201–208. <https://doi.org/10.1007/s12599-022-00783-7>
- [42] Tim Kollmer and Andreas Eckhardt. 2023. Insights on Attention Capture Dark Patterns in Social Networking Sites. BT - 29th Americas Conference on Information Systems, AMCIS 2023, Panama City, Panama, August 10-12, 2023. https://aisel.aisnet.org/amcis2023/social_comput/social_comput/14
- [43] Monica Kowalczyk, Johanna T Gunawan, David Choffnes, Daniel J Dubois, Woodrow Hartzog, and Christo Wilson. 2023. Understanding Dark Patterns in Home IoT Devices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–27. <https://doi.org/10.1145/3544548.3581432>

- [44] Cherie Lacey and Catherine Caudwell. 2019. Cuteness as a ‘dark pattern’ in home robots. In *2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 374–381. <https://doi.org/10.1109/HRI.2019.8673274>
- [45] Piergiorgio Ladisa, Henrik Plate, Matias Martinez, and Olivier Barais. 2023. Sok: Taxonomy of attacks on open-source software supply chains. In *2023 IEEE Symposium on Security and Privacy (SP)*. IEEE, 1509–1526. <https://doi.org/10.1109/SP46215.2023.10179304>
- [46] Jamie Luguri and Lior Jacob Strahilevitz. 2021. Shining a light on dark patterns. *Journal of Legal Analysis* 13, 1 (2021), 43–109. <https://doi.org/10.1093/jla/laaa006>
- [47] Francisco Lupiáñez-Villanueva, Alba Boluda, Francesco Bogliacino, Giovanni Liva, Lucie Lechardoy, and Teresa Rodríguez de las Heras Ballell. 2022. *Behavioural study on unfair commercial practices in the digital environment: dark patterns and manipulative personalisation*. Publications Office of the European Union.
- [48] SM Hasan Mansur, Sabiha Salma, Damilola Awofisayo, and Kevin Moran. 2023. Aidui: Toward automated recognition of dark patterns in user interfaces. In *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*. IEEE, 1958–1970.
- [49] S M Hasan Mansur and Kevin Moran. 2023. Toward Automated Tools to Support Ethical GUI Design. In *2023 IEEE/ACM 45th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion)*. IEEE, 294–298. <https://doi.org/10.1109/ICSE-Companion58688.2023.00079>
- [50] Arunesh Mathur, Gunes Acar, Michael J Friedman, Eli Lucherini, Jonathan Mayer, Marshini Chetty, and Arvind Narayanan. 2019. Dark patterns at scale: Findings from a crawl of 11K shopping websites. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 1–32. <https://doi.org/10.1145/3359183>
- [51] Thomas Mildner, Gian Luca Savino, Philip R. Doyle, Benjamin R. Cowan, and Rainer Malaka. 2023. About Engaging and Governing Strategies: A Thematic Analysis of Dark Patterns in Social Networking Services. , 192:1–192:15 pages. <https://doi.org/10.1145/3544548.3580695> arXiv:2303.00476
- [52] Dmitry Nazarov and Yerkebulan Baimukhambetov. 2022. Clustering of Dark Patterns in the User Interfaces of Websites and Online Trading Portals (E-Commerce). *Mathematics* 10, 18 (2022), 3219.
- [53] Trung Tin Nguyen, Michael Backes, and Ben Stock. 2022. Freely given consent? Studying consent notice of third-party tracking and its violations of GDPR in android apps. In *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*. 2369–2383. <https://doi.org/10.1145/3548606.3560564>
- [54] Liming Nie, Kabir Sulaiman Said, Lingfei Ma, Yaowen Zheng, and Yangyang Zhao. 2023. A systematic mapping study for graphical user interface testing on mobile apps. *IET Software* (2023), 249–267. <https://doi.org/10.1049/sfw2.12123>
- [55] OECD. 2022. *Dark commercial patterns*.
- [56] Daniel M Oppenheimer. 2006. Consequences of erudite vernacular utilized irrespective of necessity: Problems with using long words needlessly. *Applied Cognitive Psychology: The Official Journal of the Society for Applied Research in Memory and Cognition* 20, 2 (2006), 139–156. <https://doi.org/10.1002/acp.1178>
- [57] Lorenzo Porcelli, Massimo Ficco, and Francesco Palmieri. 2023. Mitigating User Exposure to Dark Patterns in Cookie Banners Through Automated Consent. , 145–159 pages.
- [58] S Hrushikesava Raju, Saiyed Faiyaz Waris, S Adinarayna, Vijaya Chandra Jadala, and G Subba Rao. 2022. Smart dark pattern detection: Making aware of misleading patterns through the intended app. In *Sentimental Analysis and Deep Learning: Proceedings of ICSADL 2021*. Springer, 933–947. https://doi.org/10.1007/978-981-16-5157-1_72
- [59] Brennan Schaffner, Neha A Lingareddy, and Marshini Chetty. 2022. Understanding account deletion and relevant dark patterns on social media. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–43.
- [60] Than Htut Soe, Cristiana Teixeira Santos, and Marija Slavkovik. 2022. Automated detection of dark patterns in cookie banners: how to do it poorly and why it is hard to do it any other way. *arXiv preprint arXiv:2204.11836* (2022). <https://doi.org/10.48550/arXiv.2204.11836> arXiv:2204.11836
- [61] José P Zagal, Staffan Björk, and Chris Lewis. 2013. Dark patterns in the design of games. In *Foundations of Digital Games 2013*.

Received 2023-09-28; accepted 2024-01-23